

RANI SANKANUR

9380744549 | ranisankanur@gmail.com | github.com/rani-a-s | Bagalkot

PROFESSIONAL SUMMARY

Computer Science and Engineering student with a strong foundation in software development and hands-on experience building web applications and AI-driven projects. Interested in developing practical, user-focused solutions by combining programming, web technologies, databases, and applied AI techniques.

EDUCATION

Jain College of Engineering and Research, Belagavi, Karnataka	Expected 2027
B.E. in Computer Science and Engineering CGPA: 8.8/10	
Spandana PU Science College, Ilkal	MARCH 2023
Higher Secondary (12th) 90.33%	
MGHPS, Gudur (SC)	JULY 2021
SSLC (10th) 91.84%	

PROJECTS

AI Hallucination Mitigation System	Python, RAG, SciBERT, FAISS, FastAPI
- Developed an evidence-grounded AI verification system that retrieves relevant sources and validates generated claims against supporting evidence. - Implemented multi-source retrieval and claim verification using SciFact, PubMed, and Wikipedia, classifying claims as supported, refuted, or uncertain. - Trained and evaluated a SciBERT-based verifier, achieving 81.77% validation accuracy in the current training run. - Added evidence traceability and confidence scoring with REST APIs using FastAPI for explainable AI verification.	
Harmony Heal	HTML5, JavaScript, PHP, MySQL, Apache, WAMP
- Developed a music-based wellness application that recommends relaxation, focus, and stress-relief music based on user preferences. - Implemented a music therapy engine with mood-based filtering, music database management, and customizable session timers. - Designed an interactive audio player supporting play/pause/stop, volume control, and loop functionality. - Incorporated a user feedback module to capture relaxation ratings and mood improvement, generating wellness scores and recommendations for subsequent sessions.	
CodePilot – AI Software Engineering Agent	Python, FastAPI, React.js, JavaScript, HTML5, CSS3, Sentence Transformers, NumPy, Pytest
- Developed an AI-powered repository analysis system that identifies files, directories, file types, project statistics, programming languages, frameworks, and dependencies. - Implemented AST-based Python code analysis to extract classes, functions, and imports, and created searchable code chunks with file names, symbols, types, and line numbers. - Built keyword-based and semantic code search using Sentence Transformers and embeddings to retrieve relevant code from natural-language queries. - Validated application functionality with automated testing using Pytest.	

TECHNICAL SKILLS

Programming: Python, JavaScript, SQL

Web Development: HTML5, CSS3, React.js, PHP

AI/ML: Machine Learning, NLP, RAG, SciBERT, Sentence Transformers

Backend: FastAPI, REST APIs

Databases: MySQL, MongoDB

Tools: Git, GitHub, FAISS, WAMP, Apache, Pytest, NumPy

CERTIFICATIONS

- Python – Delithe
- MySQL and DBMS – Infosys Springboard
- Cryptography and Cybersecurity – Sanfoundry

ACHIEVEMENTS

- Developed multiple end-to-end AI and software engineering projects, demonstrating practical problem-solving and development skills.
- Designed and implemented an AI hallucination detection and mitigation project to improve the reliability of AI-generated responses.
- Built and deployed academic projects integrating web technologies, databases, and AI concepts.